



Using formulas in trigonometry

TRIGONOMETRY FORMULAS CAN BE USED TO WORK OUT THE LENGTHS OF SIDES AND SIZES OF ANGLES IN TRIANGLES.

SEE ALSO

◀ 56–59 Ratio and proportion

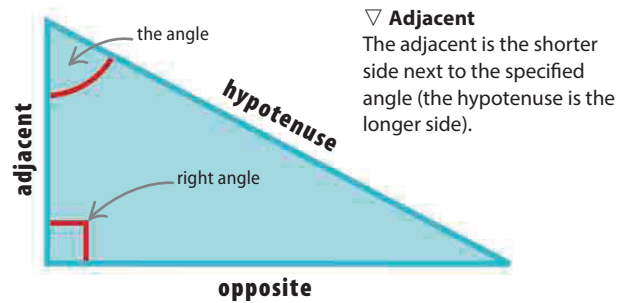
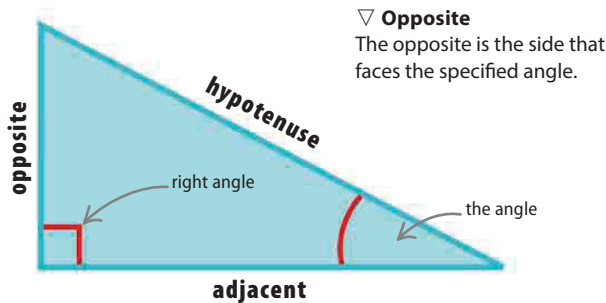
◀ 125–127 Similar triangles

Finding missing sides **162–163**

Finding missing angles **164–165**

Right triangles

The sides of these triangles are called the hypotenuse, opposite, and adjacent. The hypotenuse is always the side opposite the right angle. The names of the other two sides depend on where they are in relation to the particular angle specified.



Trigonometry formulas

There are three basic formulas used in trigonometry. "A" stands in for the angle that is being found (this may also sometimes be written as θ). The formula to use depends on the sides of the triangle that are known.

$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$$

△ The sine formula

The sine formula is used when the lengths of the opposite and hypotenuse are known.

$$\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

△ The cosine formula

The cosine formula is used when the lengths of the adjacent and hypotenuse are known.

$$\tan A = \frac{\text{opposite}}{\text{adjacent}}$$

△ The tangent formula

The tangent formula is used when the lengths of the opposite and adjacent are known.

Using a calculator

The values of sine, cosine, and tangent are set for each angle. Calculators have buttons that retrieve these values. Use them to find the sine, cosine, or tangent of a particular angle.



△ Sine, cosine, and tangent

Press the sine, cosine, or tangent button then enter the value of the angle to find its sine, cosine, or tangent.



△ Inverse sine, cosine, and tangent

Press the shift button, then the sin, cosine, or tangent button, then enter the value of the sine, cosine, or tangent to find the inverse (the angle in degrees).